

Defining the determinant

Rmk

$\det(A^T) = \det(A)$ tells us that the det has the same properties w.r.t column operations as it does w.r.t row operations

Explicitly:

$$(1) \det([v_1, v_2, \dots, kv_i, \dots, v_n])$$

$$= k \cdot \det([v_1, v_2, \dots, v_n])$$

$$(2) \det([v_1, v_2, \dots, v_i + kv_j, \dots, v_j, \dots, v_n])$$

$$= \det([v_1, v_2, \dots, v_n])$$

$$(3) \det([v_1, v_2, \dots, v_i, \dots, v_j, \dots, v_n])$$

$$= - \det([v_1, v_2, \dots, v_i, \dots, v_i, \dots, v_n])$$

3 x 3 determinant

$$\det \begin{pmatrix} \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ \underbrace{a_{3,1}}_{v_1} & \underbrace{a_{3,2}}_{v_2} & \underbrace{a_{3,3}}_{v_3} \end{bmatrix} \end{pmatrix}$$

$$= \det([v_1, v_2, v_3])$$

$$v_1 = a_{1,1}e_1 + a_{2,1}e_2 + a_{3,1}e_3$$

$$\det([a_{1,1}e_1 + a_{2,1}e_2 + a_{3,1}e_3, v_2, v_3])$$

$$= \boxed{a_{1,1} \det([e_1, v_2, v_3])} \\ + a_{2,1} \det([e_2, v_2, v_3]) \\ + a_{3,1} \det([e_3, v_2, v_3])$$

$$\rightarrow a_{1,1} \det([e_1, a_{1,2}e_1 + a_{2,2}e_2 + a_{3,2}e_3, v_3])$$

$$= a_{1,1} a_{1,2} \det([e_1, e_1, v_3])$$

$$+ a_{1,1} a_{2,2} \det([e_1, e_2, v_3])$$

$$+ a_{1,1} a_{3,2} \det([e_1, e_3, v_3])$$

$$\rightarrow a_{1,1} a_{2,2} \det([e_1, e_2, a_{1,3}e_1 + a_{2,3}e_2 + a_{3,3}e_3])$$

$$= a_{1,1} a_{2,2} a_{1,3} \det([e_1, e_2, e_1])$$

$$+ a_{1,1} a_{2,2} a_{2,3} \det([e_1, e_2, e_2])$$

$$+ a_{1,1} a_{2,2} a_{3,3} \det([e_1, e_2, e_3])$$

$$= a_{1,1} a_{2,2} a_{3,3}$$

$$\rightarrow a_{1,1} a_{3,2} a_{2,3} \det([e_1, e_3, e_2])$$

$$= -a_{1,1} a_{3,2} a_{2,3}$$

In fact, quite generally, this process tells me

$$\det([v_1, v_2, v_3])$$

$$\equiv$$

$$\sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 a_{i,1} a_{j,2} a_{k,3} \det([e_i, e_j, e_k])$$

$\neq 0$
if & only if
 i, j, k are all
distinct

There are 6 such possibilities for
 (i, j, k) all distinct

$$\bullet (1, 2, 3)$$

$$\bullet (1, 3, 2)$$

$$\bullet (2, 1, 3)$$

$$\bullet (2, 3, 1)$$

$$\bullet (3, 1, 2)$$

$$\bullet (3, 2, 1)$$

So in fact

$$\det([v_1, v_2, v_3])$$

$$= \underline{a_{1,1}} \underline{a_{2,2}} a_{3,3} \det([e_1, \cancel{e_2}, e_3])$$

$$+ a_{1,1} \underline{a_{3,2}} a_{2,3} \det([e_1, \cancel{e_3}, e_2])$$

$$+ \underline{a_{2,1}} \underline{a_{1,2}} a_{3,3} \det([e_2, \cancel{e_1}, e_3])$$

$$+ a_{2,1} \underline{a_{3,2}} a_{1,3} \det([e_2 \ e_3 \ e_1])$$

$$+ \underline{a_{3,1}} \underline{a_{1,2}} a_{2,3} \det([e_3 \ e_1 \ e_2])$$

$$- a_{3,1} \underline{a_{2,2}} a_{1,3} \det([e_3 \ e_2 \ e_1])$$

3x3-determinant

$$\det \begin{pmatrix} \underline{a_{1,1}} & \underline{a_{1,2}} & \underline{a_{1,3}} \\ \underline{a_{2,1}} & \underline{a_{2,2}} & \underline{a_{2,3}} \\ \underline{a_{3,1}} & \underline{a_{3,2}} & \underline{a_{3,3}} \end{pmatrix}$$

$$\begin{aligned} &= a_{1,1} a_{2,2} a_{3,3} - a_{1,1} a_{2,3} a_{3,2} \\ &\quad - a_{2,1} a_{1,2} a_{3,3} + a_{2,1} a_{3,2} a_{1,3} \\ &\quad + a_{3,1} a_{1,2} a_{2,3} - a_{3,1} a_{2,2} a_{1,3} \end{aligned}$$

Observation:

(Expansion by minors)

$$\det = a_{1,1} \det \begin{pmatrix} a_{2,2} & a_{2,3} \\ a_{3,2} & a_{3,3} \end{pmatrix}$$

$$- a_{2,1} \det \begin{pmatrix} a_{1,2} & a_{1,3} \\ a_{3,2} & a_{3,3} \end{pmatrix}$$

$$+ a_{3,1} \det \begin{pmatrix} a_{1,2} & a_{1,3} \\ a_{2,2} & a_{2,3} \end{pmatrix}$$

(2x2) - minors of the matrix.

Notation

$$\det(A) = |A|$$

Example

$$\begin{vmatrix} -8 & 4 & 4 \\ 0 & -1 & 3 \\ 1 & 2 & 1 \end{vmatrix}$$

$$= -8 \begin{vmatrix} -1 & 3 \\ 2 & 1 \end{vmatrix} - 0 \begin{vmatrix} 4 & 4 \\ 2 & 1 \end{vmatrix}$$

(Red annotations: arrow from -8 to 3 labeled -7, arrow from 0 to 4 labeled 20)

$$+ 1 \begin{vmatrix} 4 & 4 \\ -1 & 3 \end{vmatrix}$$

(Red annotations: arrow from 1 to 4 labeled 16, arrow from 1 to -1 labeled -4)

$$= 72$$

Observation

(Upper triangular determinants)

$$\begin{vmatrix} a_{1,1} & 0 & 0 \\ 0 & a_{2,2} & 0 \\ 0 & 0 & a_{3,3} \end{vmatrix}$$

$$= a_{1,1} \begin{vmatrix} a_{2,2} & 0 \\ 0 & a_{3,3} \end{vmatrix}$$

$$= a_{1,1} a_{2,2} a_{3,3}$$

Defⁿ (Expansion by minors)

$$\det \begin{pmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & - & - & a_{2,n} \\ \vdots & & & \\ a_{n,1} & - & - & a_{n,n} \end{pmatrix}$$

$$\begin{aligned} = & a_{1,1} |A_{1,1}| - a_{2,1} |A_{2,1}| \\ & + a_{3,1} |A_{3,1}| + \dots + (-1)^{i+1} a_{i,1} |A_{i,1}| \\ & + \dots + (-1)^{n+1} a_{n,1} |A_{n,1}| \end{aligned}$$

(n-1) x (n-1) minors

where $A_{i,j}$ is the $(n-1) \times (n-1)$ -matrix
obtained by omitting the i -th row
& j -th column

